

Where We Stand in the Race to Quantum Computing Technology

A quantum computer is a machine that performs calculations based on the laws of quantum mechanics. The excitement surrounding quantum computing is mainly due to the fact that it reduces the computational complexities of certain problems.



There's a race to build a machine that can outperform a classical one. In the current industrial or commercial domain, we have no quantum machine that can outperform a classical one. The goal, however, is to build a quantum machine that can solve problems faster than classical computers. This is called quantum supremacy or quantum ascendancy.

The below-mentioned technologies are considered appropriate and relevant based on the present state of the art. These technologies are only illustrative and

Quantum entanglement is a process by which microscopic objects like electrons or atoms lose their individuality to become better coordinated with each other. Entanglement is at the heart of new quantum technologies that can lead to great advances not only in computing but also in communication and sensing.

These entangled states are very fragile; even a tiny disturbance will undo the entanglement. Hence, it is important for current quantum technologies to physically isolate the microscopic systems they work with. These typically operate at temperatures close to absolute zero or -273 degrees Celsius.

What are entangled states? Pairs or groups of particles that are entangled can become so intertwined that measuring the properties (through

the state) of one will instantly change the properties of others. Quantum computers must have several dozen entangled states or qubits to become a commercially viable computing method.

A quantum computer is a machine that performs calculations based on the laws of quantum mechanics, which is the behaviour of particles at the sub-atomic level. The excitement surrounding quantum computing is mainly due to the fact that it reduces the computational complexities of certain problems. Only very special classes of algorithms are known to enjoy quantum speedup. Based on the evaluation of publicly available information regarding progress to date in the field of quantum computing, there appears no fundamental reason why a large, fault-tolerant quantum computer could not be built in principle.

by no means complete. Significant technical challenges remain on the path to build such technologies, and to deploy these to practical advantage.

Error-correction technology

Currently, the most powerful quantum computers are 'noisy intermediate-scale quantum' (NISQ) devices that are very sensitive to errors. Error in preparation as well as measurement of qubits is difficult to quantify. For example, for 100 qubits, a one per cent measurement error means that a NISQ device will produce an incorrect answer about 63 per cent of the time.

Eric Hudson, professor of astronomy at the University of California, Los Angeles, and his colleagues recently developed a new qubit hosted in a laser-cooled, radioactive barium ion. This 'goldilocks ion' has nearly ideal properties for realising ultra-low error

